Project Title

FlowMax: Network Flow Optimization System

Data Structures and Algorithms -3 (25CS2103E)

in

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ABSTRACT

The FlowMax: Network Flow Optimization System is a Data Structures and Algorithms project designed to determine the maximum possible flow from a source vertex to a destination vertex

in a directed network with limited edge capacities. The system uses the Edmonds-Karp algorithm, which is based on the Ford-Fulkerson method and utilizes Breadth-First Search (BFS) to identify augmenting paths efficiently. It maintains a residual network, calculates bottleneck capacities, and updates forward and reverse edge capacities after each iteration while ensuring flow conservation at intermediate vertices. The system continues this process until no further augmenting path is available, thereby determining the maximum flow. It also identifies the minimum cut and demonstrates the Max-Flow Min-Cut theorem, which establishes the relationship between maximum flow and minimum cut capacity. This project provides a practical understanding of graph representation, queues, residual networks, and network optimization techniques. FlowMax can be applied to real-world problems such as communication bandwidth allocation, traffic management, pipeline distribution, logistics, and resource allocation. Overall, the project offers an efficient and systematic approach to solving maximum-flow problems while improving the understanding of graph algorithms and their practical applications.

Signature of the Guide